

Linux File Permissions & Security Management

A practical reference for managing Linux file access, ownership, and basic security.

1. Linux Permission Model

Linux uses a permission-based security model to control access to files and directories. Every file and directory has an owner, a group, and permissions that determine who can read, write, or execute it.

User (u) — file owner

Group (g) — users belonging to the file's group

Others (o) — all other users

2. Understanding File Permissions

Use `ls -l` to view permissions, ownership, and file details.

```
ls -l
```

```
-rw-r--r-- 1 user user 1200 Jun 23 10:00 file.txt
```

Permission breakdown:

```
-rw-r--r--
| | |
| | | Others: read only
| | | Group: read only
| | | User: read and write
| | | File type (- = file, d = directory)
```

3. Permission Types

Permission	Symbol	Value
Read	r	4
Write	w	2
Execute	x	1

4. Numeric Permission Values

Numeric permissions combine read (4), write (2), and execute (1) for each permission class.

Permission	Value
rwX	7
rw-	6
r-X	5
r--	4

Permission	Value
--X	1
---	0

```
chmod 755 file.sh
```

755 = Owner rwx (7), Group r-x (5), Others r-x (5).

5. chmod — Change Permissions

Use chmod to change file or directory permissions.

Give execute permission

```
chmod +x script.sh
ls -l script.sh
```

Adds execute permission; verify with ls -l.

Remove write permission

```
chmod -w file.txt
```

Removes write permission from user, group, and others when no class is specified.

Set permissions using numbers

```
chmod 644 file.txt
```

Owner = read/write; Group = read; Others = read.

Set permissions to 755

```
chmod 755 script.sh
```

A common permission setting for executable scripts.

6. chown — Change Ownership

Use chown to change the ownership of files and directories.

Change file owner

```
sudo chown alice file.txt
ls -l file.txt
```

Changes the owner to alice.

Change owner and group

```
sudo chown alice:developers file.txt
```

Changes both the owner and group.

7. chgrp — Change Group Ownership

Use chgrp to change the group ownership of files and directories.

```
sudo chgrp developers file.txt
ls -l file.txt
```

Assigns the file to the developers group.

8. Directory Permissions

Directory permissions determine who can access the directory. A mode such as 700 allows only the owner to access it.

```
mkdir project
chmod 700 project
ls -ld project
```

700 produces drwx-----: only the owner has access.

9. Basic Security Measures

Restrict access to sensitive files

```
chmod 600 confidential.txt
```

Only the owner can read and write the file.

Restrict access to scripts

```
chmod 700 backup.sh
```

Only the owner has read, write, and execute permissions.

Secure a directory

```
chmod 700 private_data
```

Only the owner can access the directory.

Find world-writable files

```
find /home -type f -perm -002
```

Identifies files writable by everyone, which may pose security risks.

Check file ownership

```
ls -l
```

Helps verify correct ownership and permissions.

10. Common Permission Settings

Permission	Meaning
777	Full access to everyone (generally not recommended).
755	Owner full access; group and others read and execute.
744	Owner full access; group and others read only.
700	Only the owner has access.
644	Owner read/write; group and others read.
600	Owner read/write only.

11. Quick Command Reference

Command	Purpose
ls -l	View permissions, ownership, and file details.

Command	Purpose
chmod	Change file or directory permissions.
chown	Change file owner and optionally group.
chgrp	Change group ownership.
find /home -type f -perm -002	Find files writable by everyone.

Key Takeaways

- Permissions are organized around user, group, and others.
- chmod manages permissions; chown manages ownership; chgrp manages group ownership.
- Use restrictive permissions for sensitive files, scripts, and directories where appropriate.
- Review broad permissions such as 777 and world-writable files carefully.

Conclusion

Linux file security relies on correctly managing permissions and ownership. Using chmod, chown, and chgrp helps control access to files and directories and reduces the risk of unauthorized access.